

Types of functions. Part 1

In this session we will revisit the types of functions and their main characteristics, such as domain, range, asymptotes, and holes¹. Furthermore, we will also review the translation of functions. This is because by knowing the domain and range of these types of functions is going to help in identifying the domain and range of the derivatives.

Polynomials For polynomials of the form²:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0$$

where n is a nonnegative integer.

Polynomial Type	Domain	Range
Odd degree (n odd)	$(-\infty, \infty)$	$(-\infty, \infty)$
Even degree (n even)	$(-\infty, \infty)$	Depends on leading coefficient:
$a_n > 0$		$[k, \infty)$
$a_n < 0$		$(-\infty, k]$

Exponential Functions For functions of the form³:

$$f(x) = a^x \quad \text{or} \quad f(x) = b \cdot a^{kx} + c$$

where $a > 0$, $a \neq 1$, and $b \neq 0$.

Function Type	Domain	Range
Basic: $f(x) = a^x$	$(-\infty, \infty)$	$(0, \infty)$
With vertical shift: $f(x) = a^x + c$	$(-\infty, \infty)$	(c, ∞)
General: $f(x) = b \cdot a^{kx} + c$	$(-\infty, \infty)$	If $b > 0$: (c, ∞) If $b < 0$: $(-\infty, c)$

¹ January 16

² Notes:

- Domain of *all* polynomials is all real numbers, $\mathbb{R} = (-\infty, \infty)$.
- For odd-degree polynomials, the range is always all real numbers due to the end behavior: as $x \rightarrow \pm\infty$, $f(x) \rightarrow \pm\infty$ (or $\mp\infty$ if $a_n < 0$).
- For even-degree polynomials:
 - If $a_n > 0$, the graph opens upward with a **minimum value** k .
 - If $a_n < 0$, the graph opens downward with a **maximum value** k .
 - k is found using calculus techniques (derivative = 0) or vertex formulas for quadratics.

³ Special Cases

- **Natural exponential:** $f(x) = e^x$ has domain $\mathbb{R}$ and range $(0, \infty)$.
- **Natural logarithm:** $f(x) = \ln(x)$ has domain $(0, \infty)$ and range $\mathbb{R}$.
- **Exponential with base** $0 < a < 1$: Domain is still $\mathbb{R}$, range is $(0, \infty)$ if no vertical shift.
- **Key relationship:** $f(x) = a^x$ and $g(x) = \log_a(x)$ are inverse functions:
 - Domain of f = Range of $g = \mathbb{R}$
 - Range of f = Domain of $g = (0, \infty)$

Calculus Connection

- For $f(x) = a^x$, the derivative is $f'(x) = a^x \ln(a)$
- For $f(x) = \ln|x|$, the derivative is $f'(x) = \frac{1}{x}$ (for $x \neq 0$)
- Domain restrictions are crucial when finding derivatives: $\frac{d}{dx} [\ln(u(x))] = \frac{u'(x)}{u(x)}$ only when $u(x) > 0$